

TEXNIK TOPSHIRIQ (TZ)

EduCRM

O'quv markazlari uchun CRM/LMS

Versiya 2.0 · Senior review: optimallashtirilgan va kengaytirilgan

Versiya	2.0 (v1.0 — 16.09.2026 o'rniga)
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0. HUJJAT HAQIDA

0.1 Maqsad

Ushbu hujjat o'quv markazlari uchun CRM/LMS platformasi — **EduCRM** ning funksional va nofunksional talablarini belgilaydi. Hujjat shunday darajada batafsilki, uni o'qigan senior engineer qo'shimcha savolsiz implementatsiyani boshlay oladi. Ekranlar va foydalanuvchi bilan ishlash tartibi alohida hujjatda — **UX Spetsifikatsiyasi v2.0** da.

0.2 Auditoriya

Backend/frontend dasturchilar, QA, DevOps, product owner, loyiha menejeri.

0.3 Manba va referens

Funksional qamrovni to'liq tushunish uchun Hollihops CRM ning **ochiq hujjatlari** (hollipedia) tahlil qilingan. Kod, UI dizayn, matn va brend elementlari ko'chirilmaydi. Funksionallik va biznes-logika — ochiq domen bilimi.

0.4 Terminologiya konvetsiyasi

- **SHART (MUST)** — majburiy talab
- **KERAK (SHOULD)** — kuchli tavsiya, chetlanish asoslanishi shart
- **MUMKIN (MAY)** — ixtiyoriy
- Texnik atamalar (endpoint, migration, constraint, idempotency) ingliz tilida qoldiriladi.
- v1.0 dagi aralash ruscha atamalar olib tashlandi: *списание* → **yechim (charge)**, *излишки* → **hisobdan chiqarish (write-off)**, *отработка* → **qayta dars (makeup)**.

0.5 v2.0 da nima o'zgardi (senior review natijasi)

v1.0 kuchli poydevorga ega edi. v2.0 da topilgan **texnik xatolar tuzatildi**, **qarama-qarshiliklar** bartaraf etildi va **biznes qiymati yuqori** bo'lgan funksiyalar qo'shildi.

#	O'zgarish	Turi	Bo'lim
1	Charge idempotency indeksi storno'dan keyin qayta yechim qo'yishni bloklardi — lesson_charges proyeksiya jadvali bilan tuzatildi	Kritik xato	M6.6
2	CREATE RULE ... DO INSTEAD NOTHING xatoni jimgina yutib yuboradi — RAISE EXCEPTION trigger bilan almashtirildi	Kritik xato	M6.4
3	RLS + Prisma connection pool: SET LOCAL faqat transaction ichida ishlaydi; FORCE RLS va alohida DB rol qo'shildi	Kritik xato	5.3
4	moved statusdagi dars konflikt constraint'ida hisobga olinardi — tuzatildi	Xato	M4.4
5	provider_transaction_id global unique edi — (provider, provider_transaction_id) ga o'zgartirildi	Xato	M6.7
6	Payme (JSON-RPC, Basic auth) va Click (Prepare/Complete, sign_string) protokollari aniq yozildi	Aniqlashtirish	M6.7
7	MVP qamrovi va roadmap o'rtasidagi ziddiyat — yagona Release R1/R2/R3 modeli	Qarama-qarshilik	1.3, 12
8	UTM / ariza (inbound) ma'lumot modeli yo'q edi — inbound_submissions qo'shildi	Yetishmovchilik	M2.2
9	BR-A3 (7 kun) va BR-A5 (24 soat) qoidalari ziddiyatli edi — yagona "davomat oynasi" modeli	Qarama-qarshilik	M5.3
10	Dars avtomatik yakunlanishi aniqlanmagan edi — lesson auto-complete qoidasi	Yetishmovchilik	M4.7
11	Web uchun JWT saqlash usuli noaniq — httpOnly cookie + BFF	Xavfsizlik	8.3
12	12 ta yangi funksiya (muzlatish, navbat, oilaviy hisob, churn-risk, bo'lib to'lash, sertifikat va boshq.)	Yangi	M12

Belgilar: yashil blok — **yangi funksiya**, qizil blok — **tuzatilgan xato yoki kritik talab**, sariq blok — **qaror kutilmoqda**.

1. LOYIHA UMUMIY TAVSIFI

1.1 Muammo bayoni

O'quv markazlari quyidagi jarayonlarni bir-biriga bog'lanmagan holda (Excel, Telegram, qog'oz jurnal, alohida buxgalteriya) yuritadi:

1. Lead kelib tushishi va sotuv voronkasi
2. Guruh shakllantirish, jadval, xona va o'qituvchi taqsimoti
3. Davomat va uning pul hisob-kitobiga ta'siri
4. Abonement, to'lov, qarzdorlik, shaxsiy hisob
5. O'qituvchi ish haqi
6. Ota-ona/o'quvchi bilan kommunikatsiya
7. Menejment analitikasi (retention, LTV, guruh rentabelligi)

Natija: **pul yo'qoladi** (hisoblanmagan darslar, unutilgan qarzdorlar), **vaqt yo'qoladi** (qo'lda maosh hisoblash), **mijoz yo'qoladi** (lead bilan ishlanmagan, ketish xavfi o'z vaqtida sezilmagan).

1.2 Maqsadlar va o'lchov mezonlari

#	Maqsad	O'lchov mezonlari (pilotdan 3 oy keyin)
G1	Lead'dan to'lovgacha yo'lni bitta tizimda yopish	Lead → student konversiyasi 100% kuzatiladi; javobsiz lead ulushi < 5%
G2	Moliyaviy hisobni avtomatlashtirish	Yechimlarning 100% i avtomatik; qo'lda tuzatish < 1%
G3	O'qituvchi maoshini avtomatik hisoblash	Oylik payroll vaqti 1 kundan 10 daqiqagacha
G4	Ota-ona uchun shaffoflik	Faol ota-onalarning 60%+ i kabinet/botdan foydalanadi
G5	Qarzdorlikni kamaytirish	30+ kunlik qarz summasi pilotdan oldingiga nisbatan –30%
G6	Retention'ni oshirish	Oylik churn –15% (churn-risk ogohlantirishlari orqali)

1.3 Qamrov — Release modeli

v1.0 da "MVP = 11 modul" deyilgan, lekin roadmap "MVP'ni 1-2-5 bosqich bilan cheklang" degan edi. Bu ziddiyat. v2.0 da yagona model:

Release	Mazmun	Maqsad
R1 — Core MVP	IAM/RBAC, filial/xona/xodim, ma'lumotnomalar, o'quvchi, o'quv birligi, jadval + konflikt, davomat, narx, ledger, yechim dvigateli, to'lov (qo'lda + Payme/Click), qarzdorlar, ota-ona kabineti (PWA), o'qituvchi mobil davomati, Telegram/SMS asosiy xabarnomalar, import, audit	Pul to'g'ri hisoblanadi, administrator Excel'dan voz kechadi
R2 — Growth	Lead/voronka, inbound + UTM, sinov darsi, vazifalar, auto-actions, payroll, ommaviy rassilka, omnichannel chat, churn-risk, navbat (waitlist), bo'lib to'lash	Sotuv va retention avtomatlashadi
R3 — Scale	Analitika va KPI dashboard, rentabellik, sertifikatlar, so'rovnomalar (NPS), public API + outgoing webhooks, multi-tenant SaaS onboarding, Uzum Bank, telefoniya	SaaS sifatida sotishga tayyor

Har bir release o'zi alohida ishlaydigan mahsulot. R1 pilot filialda real ishlatilmaguncha R2 boshlanmaydi — real foydalanish R2 ustuvorliklarini o'zgartiradi.

1.4 Qamrovdan tashqari (barcha release'lar)

- To'liq LMS (video hosting, interaktiv mashqlar, avto-tekshiriladigan testlar)
- Native mobil ilova (PWA + API tayyor)
- 1C va soliq hisoboti integratsiyasi; onlayn kassa / fiskal chek (aloqida huquqiy tahlil)
- Kutubxona, ekskursiya, imtihon moduli, loyalty ball, korporativ (B2B) bo'lim

1.5 Mahsulot strategiyasi

Birinchi bosqich — bitta tashkilot (pilot: Najot Ta'lim). Ma'lumot modeli **multi-tenant SaaS'ga tayyor**: har bir jadvalda `tenant_id`, PostgreSQL RLS, tenant-scoped kesh kalitlari va fayl yo'llari. Bu MVP'ni murakkablashtirmaydi, keyin esa schema'ni qayta yozishdan qutqaradi.

2. GLOSSARIY (DOMAIN TILI)

Barcha qatlamlarda (DB, API, UI, hujjat) bir xil atama ishlatiladi.

Atama (uz)	Kod (en)	Ta'rif
Tenant	tenant	Tizimdan foydalanuvchi tashkilot
Filial	branch	Tashkilotning jismoniy bo'linmasi
Xona	classroom	Filialdagi o'quv xonasi
Xodim	employee	Tizimga kiruvchi ichki foydalanuvchi
O'qituvchi	teacher	Dars o'tuvchi xodim (employee profili)
Mijoz	client	Jismoniy shaxs (bir odam)
Oila	household	Bir to'lovchi ostidagi mijozlar guruhi (YANGI)
Ariza	inbound_submission	Tashqi manbadan kelgan xom so'rov (forma, bot) — UTM bilan (YANGI)
Lead	lead	Bitta kursni sotish imkoniyati, odam emas
O'quvchi	student	Sotuvdan keyingi o'quv kartochkasi; bir mijozda bitta
Aloqa shaxsi	contact_person	Ota-ona/vakil
O'quv birligi	study_unit	Guruh, mini-guruh yoki individual — darslar va pul konteyneri
A'zolik	enrollment	O'quvchining o'quv birligiga birlashtirilishi (sana + narx)
Muzlatish	enrollment_freeze	A'zolikni vaqtincha to'xtatish (YANGI)
Navbat	waitlist_entry	To'lgan guruhga kutish (YANGI)
Jadval qoidasi	schedule_rule	Takrorlanuvchi dars ta'rifi
Dars	lesson	Aniq sanadagi mashg'ulot
Davomat	attendance	(dars × o'quvchi) holati
Narx	price	Narx modeli (versiyalanadi)
Schyot	invoice	To'lov uchun qo'yilgan hisob
Tushum	payment	Kelgan pul (ledger'da musbat)
Yechim	charge	O'qish uchun hisoblangan summa (ledger'da manfiy)
Shaxsiy hisob	ledger	Barcha moliyaviy operatsiyalar (append-only)
Balans	balance	SUM(ledger_entries.amount); manfiy = qarz
To'lov grafigi	installment_plan	Bo'lib to'lash rejasi (YANGI)
Auto-action	automation	Trigger → shart → harakat qoidasi

Modellashtirish qarori: lead — odam emas, sotuv imkoniyati. Bir ota-ona ikki farzandini turli kurslarga yozdirishi yoki bitta o'quvchi ingliz tilidan keyin IT kursiga kelishi — alohida sotuv sikllari. Lead = odam qilinsa, voronka hisoboti buziladi.

3. ROLLAR VA RUXSATLAR

3.1 Rollar

Rol	Tavsif	Tipik doira
super_admin	Platforma admini (SaaS)	Barcha tenant'lar; tenant ma'lumotiga faqat "support session" orqali, audit bilan
owner	Tashkilot egasi	Hammasi
branch_manager	Filial boshqaruvchisi	O'z filiali: to'liq
sales_manager	Sotuv menejeri	Lead, ariza, vazifa, sinov darsi
administrator	Reception	Jadval, davomat, to'lov qabuli
accountant	Buxgalter	Moliya, hisobot, payroll
teacher	O'qituvchi	O'z darslari, davomat, o'z maoshi
student	O'quvchi	O'z kabinet
parent	Ota-ona	Farzand(lar) kabinet

Rollar **shablon** hisoblanadi: tenant ularni nusxalab, o'z rolini yarata oladi (`role.clone`).

3.2 Ruxsat modeli

RBAC + branch scoping + ownership scoping. Ruxsat — `<resource>.<action>` satri, har biri `scope` bilan: `own | branch | all`.

```
branch.view|create|update|archive      classroom.manage
employee.view|create|update|deactivate role.manage
teacher.view_salary_settings|edit_salary_settings
inbound.view|process|archive|purge
lead.view|create|update|delete|view_contacts|bulk_action|change_owner
student.view|create|update|archive|merge|view_contacts|view_passport|export_personal_data
study_unit.view|create|update|delete|change_status|manage_members
waitlist.manage                       enrollment.freeze
schedule.view|create|update|delete
lesson.cancel|reschedule|substitute_teacher|complete
attendance.view|mark|mark_past|edit_locked
price.view|manage                     discount.manage
finance.view_amounts                  -- summalarni ko'rish (alohida!)
invoice.view|create|update|cancel
payment.view|create|update|delete|refund
installment.manage
ledger.view|adjust|transfer
payroll.view_own|view_all|calculate|approve|pay|reopen
report.finance|sales|load|profitability|manager|retention
message.send|bulk_send  template.manage  automation.manage
chat.view_all|reply      import.run|rollback  export.run
audit.view  settings.manage  api_key.manage
```

3.3 Xavfsizlik talablari

- **SHART:** Ruxsat tekshiruvi backend'da. Frontend'da tugma yashirish — UX, xavfsizlik emas.
- **SHART:** Har bir endpoint'da `@RequirePermission('...')`. Cl'da skript barcha controller route'larini skanerlaydi: guard'siz yoki `@Public()` belgisiz route — build xatosi.

- **SHART:** `finance.view_amounts` — administrator to'lov qabul qiladi, lekin umumiy moliyaviy manzarani (tushum jami, rentabellik) ko'rmaydi.
- **SHART:** `attendance.mark_past`, `payment.update/delete`, `payroll.reopen`, `ledger.adjust` — har doim audit log'ga yoziladi va **sabab maydoni majburiy**.
- **KERAK:** "To'rt ko'z" (maker-checker) — belgilangan summadan katta refund yoki ledger tuzatish ikkinchi xodim tasdig'ini talab qiladi (chegara sozlanadi).

4. FUNKSIONAL TALABLAR

M1. TASHKILOT VA SOZLAMALAR

M1.1 Filiallar

- CRUD: nom, manzil, telefon, ish vaqti, timezone (default `Asia/Tashkent`), qisqartma (avto-nomlash uchun)
- **SHART:** Filial o'chirilmaydi — `is_active=false`. Tarixiy moliyaviy yozuvlar bog'langan.

M1.2 Xonalar

- Nom, sig'im, filial, jihozlar, `is_active`. Bandlik `lessons` orqali aniqlanadi (M4.4).

M1.3 Ma'lumotnomalar (tenant-scoped CRUD)

Fan, daraja, yosh toifasi, guruh turi, reklama manbasi, o'qish maqsadi, lead statuslari (tartib + voronka bosqichi + `is_won/is_lost` bayrog'i), ketish sababi (churn reason), to'lov turi, to'lov usuli, xarajat toifasi, bayramlar.

KERAK: Har bir ma'lumotnoma yozuvi o'chirilmaydi, `is_active=false` qilinadi — aks holda eski lead/to'lovlardagi bog'lanish uziladi.

M1.4 Xodimlar

- To'liq ism, telefon, email, rol(lar), filial(lar), `is_active`
- O'qituvchi profili: fanlar, darajalar, yosh toifalari, maosh sozlamalari, **mavjudlik (availability)** — haftalik ish vaqti (bo'sh o'qituvchi qidiruvi uchun)
- **SHART:** Xodim o'chirilmaydi — deaktivatsiya. Deaktivatsiyada barcha sessiyalar va refresh token'lar bekor qilinadi.

M1.5 Maxsus maydonlar (custom fields)

- Entity: student, lead, study_unit. Turlar: text, number, date, select, multiselect, checkbox
- Saqlash: `custom_field_definitions` + `custom_data` JSONB, **GIN index**
- **SHART:** Backend `custom_data` ni definition bo'yicha validatsiya qiladi (tur, majburiylik, select variantlari).

M1.6 Onboarding ustasi

YANGI. Yangi tenant 15 daqiqada ishga tushadi: 1) filial va xonalar → 2) fanlar va darajalar → 3) narxlar → 4) xodimlarni taklif qilish (SMS/Telegram havola) → 5) Excel'dan o'quvchilarni import qilish → 6) birinchi guruh. Har qadam o'tkazib yuboriladi va keyin davom ettiriladi. SaaS sotuvida bu — konversiyaning asosiy omili.

M2. LEAD VA SOTUV (CRM) — R2

M2.1 Lead entity

Maydonlar: `client_id`, FIO, telefon(lar), email, fan, daraja, yosh toifasi, filial, mas'ul (`owner_id`), status, reklama manbasi, o'qish maqsadi, afzal jadval (ko'p qiymatli), teglar, izoh, custom fields, `inbound_submission_id`, `created_at`, `first_contact_at`, `converted_at`, `lost_reason_id`.

Biznes qoidalar:

- **BR-L1:** Bir client'da bir nechta lead (turli fan/davr)
- **BR-L2:** Lead to'lov qabul qilmaydi va shartnoma tuzmaydi; faqat sinov darsiga yoziladi
- **BR-L3:** Avtomatik harakatlar statusni orqaga qaytarmaydi; qo'lda o'zgarish `lead_status_history` ga yoziladi
- **BR-L4 (YANGI):** `is_lost` statusiga o'tishda ketish sababi majburiy — aks holda “nega yo'qotaymiz?” hisoboti bo'sh qoladi
- **BR-L5 (YANGI):** SLA — `first_contact_at` belgilanmagan lead N daqiqadan keyin (default 60) mas'ulga, 2N dan keyin rahbarga eskalatsiya

M2.2 Arizalar (inbound) va manbalar

TUZATILDI. v1.0 UX hujjatida “Arizalar” alohida bo'lim edi, TZ ma'lumot modelida esa umuman yo'q edi, UTM ham saqlanmasdi. Yechim: har bir tashqi so'rov avval xom holda `inbound_submissions` ga yoziladi, keyin qoida bo'yicha (avto yoki qo'lda) lead'ga aylanadi.

Manbalar: qo'lda kiritish, web-forma (`POST /api/v1/public/inbound`, rate-limit + captcha + honeypot), Telegram bot, import, webhook (Tilda/landing), Instagram (R3).

`inbound_submissions`: `source`, `form_id`, `raw_payload` JSONB, `utm_source/medium/campaign/content/term`, `referrer`, `landing_url`, `ip`, `status` (`new|processed|spam|archived`), `lead_id`.

- **SHART:** UTM va qo'lda tanlangan reklama manbasi **ikkalasi ham** saqlanadi. UTM — marketing byudjetini baholashning yagona ishonchli usuli.
- **SHART — dublikat:** telefon `+998XXXXXXXXX` ga normalizatsiya (`libphonenumber`). Topilsa — yangi lead mavjud client'ga bog'lanadi, xodimga ogohlantirish chiqadi.
- **KERAK:** Ko'rsatish qoidalar — call-markaz xodimi qaysi filial/manba arizalarini ko'rishi.

M2.3 Lead ro'yxati

- Filtrlar, jadval/kanban ko'rinishi, saqlangan ko'rinishlar (UX §P1)
- Ommaviy amallar: status, mas'ul, teg, xabar, eksport
- **SHART:** cursor-based pagination. `total` faqat so'ralganda va 10 000 dan oshsa taxminiy (`EXPLAIN estimate`)

M2.4 Sinov darsi

- Lead mavjud lesson'ga yoziladi; davomatda `trial` statusi
- **SHART:** Sinov sig'imni band qiladi, yechim yaratmaydi
- Natija: keldi / kelmadi / keldi va yozildi → mos trigger
- **KERAK:** Bir lead uchun maksimal sinov soni sozlanadi (default 1)

M2.5 Lead → Student konversiyasi

1. Lead `is_won` statusiga o'tadi
2. Client'da student yo'q bo'lsa — yaratiladi, bor bo'lsa — birlashtiriladi
3. Kontakt, manba, afzal jadval ko'chiriladi
4. Enrollment yaratiladi (o'quv birligi + sana + narx)
5. Lead `student_id` bilan bog'lanadi

SHART: Hammasi bitta DB transaction ichida, `Idempotency-Key` bilan.

M2.6 Vazifalar

- Matn, mas'ul, muddat, bog'liq obyekt, muhimlik, natija
- Takrorlanuvchi vazifalar — `RRULE` (cron emas: biznes foydalanuvchiga tushunarli)
- Bajarilgan vazifa natijasi majburiy (sozlanadi)

M2.7 Auto-actions dvigateli

Arxitektura: **trigger** → **condition** → **action**.

Trigger	Hodisa
inbound.received	Ariza keldi
lead.created / lead.status_changed	Lead yaratildi / status o'zgardi
lead.trial_booked / trial_attended / trial_missed	Sinov darsi hodisalari
lead.sla_breached	Javob muddati o'tdi (YANGI)
lead.converted	Studentga aylandi
student.enrolled / status_changed	Guruhga qo'shildi / status
student.absent_consecutive	Ketma-ket N marta kelmadi
student.churn_risk_high	Ketish xavfi yuqori (YANGI, M12.4)
invoice.issued / due_soon / overdue	Schyot hodisalari
payment.received / installment.overdue	To'lov / grafik bo'yicha kechikish
balance.negative	Balans chegaradan pastga tushdi
lesson.completed / lesson.cancelled	Dars hodisalari
enrollment.ending_soon / frozen / unfrozen	A'zolik hodisalari
waitlist.seat_available	Navbatdagiga joy bo'shadi (YANGI)

Shartlar: filial, fan, daraja, tur, status, summa chegarasi, teg, custom field qiymati.

Harakatlar: **change_status**, **create_task** (kechikish bilan), **send_message** (kanal zanjiri bilan), **assign_owner** (round-robin / yuklama bo'yicha / aniq), **add_tag**, **call_webhook**, **wait** (kechikish, R2+).

SHART — ishonchlilik:

- Outbox pattern: hodisa biznes transaction ichida **outbox_events** ga yoziladi; worker o'qib bajaradi
- Har bir bajarilish **automation_runs** da: trigger, shart natijasi, harakat, xato
- Retry: exponential backoff, 5 urinish, keyin dead-letter + alert
- Idempotency: **idempotency_key** = **hash(automation_id, event_id, action_index)**
- **SHART (YANGI) — tsikl himoyasi:** avtomatlashtirish yaratgan hodisa **causation_depth** ni oshiradi; chuqurlik > 3 bo'lsa zanjir to'xtatiladi (A statusni o'zgartiradi → B qaytaradi → cheksiz tsikl)
- **KERAK (YANGI) — dry-run:** qoidani yoqishdan oldin "oxirgi 30 kunda bu qoida necha marta ishlagan bo'lardi" preview

Muhandislik izohi: auto-action'ni service metodidan to'g'ridan-to'g'ri chaqirish (status o'zgartirish ichida SMS yuborish) — eng ko'p uchraydigan xato. Provayder sekin javob bersa, transaction uzoq ochiq qoladi va lock muammosi kelib chiqadi. Outbox majburiy.

M3. O'QUVCHILAR

M3.1 Kartochka bo'limlari

Sarlavha (FIO, ID, status, balans so'm + dars ekvivalenti, churn-risk belgisi), shaxsiy ma'lumot, kontaktlar va aloqa shaxslari ("schyot oluvchi", "xabar oladi" bayroqlari, ulangan Telegram), o'quv ma'lumoti, o'quv birliklari, jadval, shaxsiy hisob, to'lov grafigi, hujjatlar, testlar, tarix (timeline) va vazifalar, kabinet faolligi.

M3.2 Biznes qoidalari

- **BR-S1:** Bir client'da bitta student kartochkasi
- **BR-S2:** Student o'chirilmaydi — **archived**
- **BR-S3:** Status enrollment'lardan **hisoblanadi** (**active** | **frozen** | **finished** | **no_enrollment** | **archived**); qo'lda faqat **archived**

- **BR-S4:** Merge — ledger, enrollment, davomat, hujjatlar bitta kartochkaga ko'chadi; ikkinchisi `merged_into_id`. Merge bitta transaction, audit log'da to'liq diff, **24 soat ichida bekor qilish** mumkin (`merge_operations` jadvali orqali)

M3.3 Hujjatlar generatsiyasi

- Shablon (HTML → PDF, Gutenberg/Playwright) + o'zgaruvchilar `{{student.full_name}}`, `{{contract.number}}`...
- **SHART:** Shartnoma raqami tenant bo'yicha ketma-ket va bo'shliqsiz (`document_sequences` + `SELECT ... FOR UPDATE`)
- **SHART:** Generatsiya qilingan PDF o'zgarmas saqlanadi (hash bilan); shablon o'zgarsa ham eski hujjat o'zgarmaydi
- **KERAK (YANGI):** Ota-ona kabinetida shartnomani OTP bilan tasdiqlash (oddiy elektron imzo) — qog'oz imzosiz yozilish

M4. O'QUV BIRLIKLARI VA JADVAL

M4.1 O'quv birligi

Turi: `group | mini_group | individual`. Maydonlar: `branch_id` (SHART), `discipline_id` (SHART), `level_id`, `age_category_id`, `type_id`, `status` (`forming|active|paused|finished|cancelled`), `name/auto_name`, `capacity`, `min_size` (YANGI), `default_price_id`, `responsible_id`, `color`, `start_date`, `end_date`, `curriculum_id` (YANGI, M12.8), `version`.

- **BR-U1:** `forming` da darslar generatsiya qilinmaydi va yechim yo'q
- **BR-U2:** `forming` → `active` uchun jadval qoidasi SHART
- **BR-U3:** Sig'im oshsa — ogohlantirish (blokklanmaydi); sinov ishtirokchilari ham hisoblanadi
- **BR-U4 (YANGI):** O'quvchi soni `min_size` dan kam bo'lsa — guruh rentabelligi xavfi haqida ogohlantirish

M4.2 A'zolik (enrollment)

Maydonlar: `student_id`, `study_unit_id`, `start_date`, `end_date`, `price_id`, `discount_ids[]`, `status` (`active|frozen|finished|transferred|cancelled`), `contract_id`, `payer_client_id` (YANGI), `version`.

- **BR-E1:** Bir (`student`, `study_unit`) uchun sana oraliqlari kesishmaydi — `EXCLUDE USING gist (student_id WITH =, study_unit_id WITH =, daterange(start_date, end_date, '[]') WITH &&)`
- **BR-E2:** Transfer — eski yopiladi, yangi ochiladi; ledger'da `transfer_out/transfer_in` faqat enrollment'ga bog'langan avans ko'chirilsa
- **BR-E3:** Enrollment o'chirilishi faqat yechim va davomat bo'lmaganda; aks holda `cancelled`

M4.3 Jadval qoidasi va dars generatsiyasi

`schedule_rule`: `study_unit_id`, `days_of_week` `int[]`, `start_time`, `end_time` yoki `academic_hours_per_day`, `break_minutes`, `period_start/end` yoki `total_academic_hours`, `classroom_id`, `teacher_id`, `is_active`.

SHART: Darslar materializatsiya qilinadi. Sabab: konflikt nazorati DB constraint'da, har bir darsni alohida tahrirlash, davomat va yechim darsga bog'lanadi, har so'rovda RRULE hisoblash sekin.

Generatsiya qoidalari:

- Gorizont — **rolling 6 oy** (v1.0 da 12 oy edi): kechki job har kuni gorizontni uzaytiradi. Kamroq "yetim" dars, tezroq regeneratsiya
- Qoida o'zgarganda: o'tgan darslar tegilmaydi, kelajakdagilar regeneratsiya
- **SHART:** Davomati yoki yechimi bor dars avtomatik o'chirilmaydi
- **SHART:** Qo'lda o'zgartirilgan dars (`is_detached=true`) regeneratsiyada ustidan yozilmaydi
- Bayramlar o'tkazib yuboriladi; `total_academic_hours` rejimida davr oxiriga suriladi

- **SHART:** `POST /schedule-rules/:id/preview` — saqlashdan oldin natija va konfliktlar ro'yxati

M4.4 Konflikt nazorati (kritik)

1. Xona konflikti — bloklanadi
2. O'qituvchi konflikti — bloklanadi
3. O'quvchi konflikti — ogohlantirish

```
ALTER TABLE lessons ADD CONSTRAINT lessons_classroom_no_overlap
EXCLUDE USING gist (classroom_id WITH =, tstzrange(start_at, end_at) WITH &&)
WHERE (status NOT IN ('cancelled','moved') AND classroom_id IS NOT NULL);

ALTER TABLE lessons ADD CONSTRAINT lessons_teacher_no_overlap
EXCLUDE USING gist (actual_teacher_id WITH =, tstzrange(start_at, end_at) WITH &&)
WHERE (status NOT IN ('cancelled','moved') AND actual_teacher_id IS NOT NULL);
-- btree_gist extension talab qilinadi
```

TUZATILDI. v1.0 da `moved` statusidagi (ko'chirilgan) eski dars ham constraint'da qolgan edi — ko'chirilgandan keyin eski vaqt “band” bo'lib turardi. O'qituvchi konflikti esa `teacher_id` bo'yicha tekshirilardi — almashtirishda haqiqiy dars o'tuvchi `actual_teacher_id` (default = `teacher_id`). Ikkalasi tuzatildi.

Nega DB darajasida? `SELECT ... overlaps` keyin `INSERT` — race condition. Exclusion constraint buni to'liq yopadi. Application tekshiruvi baribir kerak — UX uchun: constraint xatosi `SCHEDULE_CONFLICT` kodiga aylantirilib, **bo'sh muqobil xona va vaqtlar** taklifi bilan qaytariladi.

M4.5 Jadval bilan amallar

Bitta darsni ko'chirish (`moved_to_lesson_id`), bekor qilish (sabab bilan, yechimsiz), o'qituvchini almashtirish (bitta dars yoki davr; payroll `actual_teacher_id` ga), xona o'zgartirish, qo'shimcha dars, qayta dars (makeup, `makeup_for_lesson_id`), **ommaviy ko'chirish** (YANGI: “o'qituvchi kasal — keyingi 3 kun darslarini bekor qil va xabar ber” bitta amalda).

M4.6 Jadval ko'rinishlari

Guruh jadvali, xonalar bo'yicha, o'qituvchilar bo'yicha, bo'sh xonalar, bo'sh o'qituvchilar (availability + konflikt + fan/daraja), interaktiv drag-and-drop kalendar.

SHART: (`study_unit_id`, `start_at`), (`classroom_id`, `start_at`), (`actual_teacher_id`, `start_at`) index'lari.

M4.7 Darsni yakunlash

TUZATILDI. v1.0 da yechim “dars `completed` bo'lganda” ishga tushardi, lekin darsni kim va qachon yakunlashi aniqlanmagan edi. Natija — yakunlanmagan darslar va hisoblanmagan pul.

- Davomat to'liq belgilangach dars avtomatik `completed`
- `end_at + 2 soat` gacha davomat belgilanmagan bo'lsa — o'qituvchiga va administratorga eslatma
- Kun oxirida (23:00) belgilanmagan darslar “Davomat kutilmoqda” ro'yxatiga tushadi (Ish stolida qizil vidjet)
- **KERAK:** Sozlama — “belgilanmagan o'quvchi = keldi” deb avtomatik yakunlash (ba'zi markazlar shunday ishlaydi), default o'chiq

M5. DAVOMAT

M5.1 Statuslar

Status	Kod	Pulga ta'siri
Keldi	present	To'liq yechim
Kech qoldi	late	To'liq yechim

Status	Kod	Pulga ta'siri
Kelmadi (sababsiz)	absent_unexcused	To'liq yechim
Kelmadi (sababli)	absent_excused	Sozlamaga qarab: to'liq / qisman % / yo'q
Muzlatilgan	frozen	Yechim yo'q (avtomatik, M12.1)
Sinov	trial	Yechim yo'q
Bepul	free	Yechim yo'q
Qayta dars	makeup	Yechim yo'q

M5.2 Davomat kartasi

Matritsa (o'quvchi × sana), katakda status + izoh + test bali, pastda dars bo'yicha jami, o'ngda o'quvchi bo'yicha qatnashish va foiz.

M5.3 Biznes qoidalar — yagona “davomat oynasi”

TUZATILDI. BR-A3 (7 kun) va BR-A5 (o'qituvchi 24 soat) bir-biriga mos kelmasdi. Endi bitta model:

Kim	Erkin tahrirlash oynasi	Oynadan keyin
O'qituvchi	Dars boshlanishidan 15 daq oldin → tugagandan 24 soat keyin	Faqat administratorga so'rov yuboradi
Administrator	7 kun (sozlanadi)	<code>attendance.mark_past</code> + sabab
Payroll davri <code>approved</code> bo'lgan dars	—	<code>attendance.edit_locked</code> + davr qayta ochiladi

- **BR-A1:** Faqat `planned/completed` darsga
- **BR-A2:** Faqat enrollment oralig'idagi o'quvchilarga
- **BR-A4:** Davomat o'zgarsa — yechim storno + qayta hisoblash (M6.6)
- **KERAK (YANGI):** O'qituvchi davomati `offline` ishlaydi (PWA, IndexedDB navbat) — signal yo'q xonada ham belgilanadi, keyin sinxronlanadi; konflikt bo'lsa server versiyasi yutadi va o'qituvchiga ko'rsatiladi

M6. MOLIYA

Modul — tizim yadrosi. Bu yerdagi xato pul yo'qotishga olib keladi va keyin tuzatish deyarli imkonsiz.

M6.1 Fundamental qoidalar (buzilmaydi)

1. Summalar **BIGINT**, **tiyinda**. FLOAT/DOUBLE/REAL hech qachon. (Payme ham tiyinda ishlaydi — konvertatsiya kerak emas.)
2. Valyuta `currency` ustunida (ISO 4217, default UZS)
3. Yaxlitlash faqat ko'rsatishda. Bo'lish natijasidagi qoldiq (masalan paketni darslarga bo'lishda) **oxirgi yozuvga qo'shiladi** — yig'indi har doim aniq
4. Ledger **append-only**; xato — storno yozuvi bilan
5. Barcha pul hisob-kitobi bitta `Money` value object orqali (TypeScript `bigint`); `number` bilan pul arifmetikasi lint qoidasi bilan taqiqlanadi

M6.2 Narx modellari

Model	Kod	Hisoblash	Qo'llanishi
Dars uchun	per_lesson	amount × dars	Kurslar
Akademik soat	per_academic_hour	amount × ak. soat	Individual
Paket	package	N dars/soat uchun jami	“40 soatlik kurs”
Oylik abonement	monthly	Oyiga belgilangan summa; to'liq bo'lmagan oy — darslar ulushi bo'yicha (pro-rata)	Najot Ta'lim tipidagi markazlar

Model	Kod	Hisoblash	Qo'llanishi
Kalendar	calendar	Kunlik stavka × kunlar	Kunlik abonement

YANGI: monthly modeli. O'zbekistondagi o'quv markazlarining aksariyati "oyiga X so'm" ishlaydi. v1.0 da bu faqat `calendar` orqali bilvosita edi. Pro-rata qoidasi: oy o'rtasida qo'shilgan o'quvchi `amount × (qolgan rejadagi darslar / oydagi rejadagi darslar)` to'laydi.

`price: name, type, amount, currency, academic_hour_minutes (45), package_units, allow_partial_payment, validity_days, branch_ids, discipline_id, level_id, valid_from/to, superseded_by_id.`

SHART: Narx o'zgarsa — yangi versiya. Enrollment eski `price_id` ga bog'langan qoladi. Narxni yangi versiyaga ommaviy o'tkazish — alohida amal, `effective_from` sanasi bilan.

M6.3 Chegirmalar

- `percent` yoki `fixed_amount`, muddat, sabab, enrollment darajasida
- Tartib: avval percent, keyin fixed; **jami chegirma narxdan oshmaydi** (manfiy narx bo'lmaydi)
- **YANGI — avtomatik chegirma qoidalari:** aka-uka (oilada 2+ faol o'quvchi), ikkinchi kurs, oldindan N oy to'lash. Qoida bajarilsa chegirma avto qo'llanadi, ko'rinadi va bekor qilinadi

M6.4 Ledger arxitekturasini

Jadval `ledger_entries` (append-only):

Ustun	Turi	Izoh
id	uuid (v7)	Vaqt bo'yicha tartiblanadi
tenant_id, student_id	uuid	
enrollment_id	uuid null	
entry_type	enum	payment, charge, refund, adjustment, bonus, write_off, transfer_in, transfer_out, reversal
amount	bigint	Ishorali: tushum +, yechim –
currency	char(3)	
occurred_at / created_at	timestamp	Biznes / texnik sana
source_type, source_id	text, uuid	lesson, invoice, manual, automation, provider
reverses_entry_id	uuid null	Storno qaysi yozuvni bekor qiladi
payment_method_id, payment_type_id, invoice_id	uuid null	
provider, provider_transaction_id	text null	Onlayn to'lov
idempotency_key	text null	UNIQUE (tenant_id, idempotency_key)
created_by, comment		

Balans: `SUM(amount) WHERE student_id=? AND currency='UZS'`. Manfiy — qarz, musbat — avans.

Performance: `students.cached_balance` bir xil transaction ichida yangilanadi (`UPDATE ... SET cached_balance = cached_balance + $1`). Har kecha reconciliation job SUM bilan solishtiradi; nomuvofiqlik — **kritik alert**.

TUZATILDI — immutability. v1.0 dagi `CREATE RULE ... DO INSTEAD NOTHING UPDATE/DELETE` ni **jimgina** yutib yuboradi: kod "muvaqqiyatli" deb o'ylaydi, ma'lumot esa o'zgarmagan — eng yomon turdagi xato. To'g'risi:

```
CREATE FUNCTION forbid_ledger_mutation() RETURNS trigger AS $$
BEGIN
    RAISE EXCEPTION 'ledger_entries is append-only (op=%)', TG_OP
    USING ERRCODE = 'P0001';
END $$ LANGUAGE plpgsql;

CREATE TRIGGER ledger_immutable
BEFORE UPDATE OR DELETE OR TRUNCATE ON ledger_entries
FOR EACH STATEMENT EXECUTE FUNCTION forbid_ledger_mutation();

-- Qo'shimcha: ilova DB roli uchun
REVOKE UPDATE, DELETE, TRUNCATE ON ledger_entries FROM app_user;
```

Nega bitta jadval? Balans ikki jadval (payments – charges) ayirmasi bo'lsa, har hisobot ikki JOIN talab qiladi va “balans tarixi” so'rovi juda murakkablashadi. Bitta append-only ledger — buxgalteriyaning standart yechimi.

M6.5 Schyot (invoice)

- `student_id, enrollment_id, payer_client_id, number, issued_at, due_at, total_amount, paid_amount, status` (`draft|issued|partially_paid|paid|cancelled`)
- **overdue** — **status emas, hisoblanadigan holat** (`due_at < now()` AND `status IN (issued, partially_paid)`). v1.0 da u status edi — bu cron'ga bog'liq va har doim kechikadigan ma'lumot
- Pozitsiyalar, qo'lda / guruhga ommaviy / avtomatik qo'yish
- To'lov schyotga taqsimlanishi: FIFO (eng eski ochiq schyotdan) yoki aniq schyotga
- **SHART:** Raqam tenant bo'yicha bo'shliqsiz (`document_sequences + FOR UPDATE`). PostgreSQL sequence emas — rollback'da bo'shliq qoladi
- **YANGI:** Har bir schyotda **qisqa to'lov havolasi** (`/p/<token>`) — Payme/Click tanlash sahifasi, login talab qilmaydi, token 30 kun amal qiladi

M6.6 Yechim dvigateli (charge engine) — kritik

Trigger: dars `completed` holatiga o'tganda yoki yakunlangan darsda davomat o'zgarganda.

```
for each attendance in lesson.attendances:
    if status in (trial, free, makeup, frozen): target = 0
    elif status == absent_excused: target = amount * settings.excused_charge_pct
    else: target = calculate_amount(enrollment.price, lesson, enrollment.discounts)

    current = lesson_charges.get(lesson.id, student.id)          -- FOR UPDATE
    if current.amount == target: continue                        -- idempotent
    if current.entry_id: insert reversal(current.entry_id)       -- storno
    if target > 0: new_id = insert charge(-target, revision+1)
    upsert lesson_charges(lesson_id, student_id, entry_id=new_id,
                          amount=target, revision=revision+1)
```

TUZATILDI — kritik xato. v1.0 dagi indeks:

`UNIQUE (source_type, source_id, student_id, entry_type) WHERE entry_type='charge' AND reverses_entry_id IS NULL`

Storno qilingan asl charge yozuvida `reverses_entry_id` NULL bo'lib qoladi (ledger o'zgarmas!).

Shuning uchun davomat o'zgargandan keyin **yangi charge qo'yib bo'lmaydi** — unique constraint xatosi. v1.0 ning o'z misoli (–50 000, +50 000, –30 000) shu indeks bilan ishlamasdi.

Yechim: `lesson_charges` — o'zgaruvchan proyeksiya jadvali: `PRIMARY KEY (lesson_id, student_id, current_entry_id, amount, revision)`. U “hozirgi holat”ni saqlaydi; ledger esa to'liq tarixni. Idempotency `revision` va `idempotency_key = hash(lesson, student, revision)` bilan ta'minlanadi.

`calculate_amount:`

- `per_lesson` → `price.amount`
- `per_academic_hour` → `price.amount × lesson.academic_hours`

- **package** → **allow_partial_payment=false** bo'lsa paket boshida bir marta; aks holda darsga bo'lib (qoldiq oxirgi darsga)
- **monthly** → oy boshida schyot, darsga yechim **amount / oydagi rejadagi darslar**
- **calendar** → alohida kunlik job, uzluksiz kunlar bitta yozuvga

M6.7 To'lov qabul qilish

- **Qo'lda:** summa, sana, usul, tur, izoh, kim qabul qildi, chek (PDF/Telegram)
- **Kassa smenasi (YANGI):** administrator naqd pul uchun smena ochadi/yopadi; yopishda tizimdagi va qo'ldagi naqd summa solishtiriladi, farq audit'ga yoziladi
- **Onlayn:** Payme, Click (R1), Uzum (R3)
- **Refund:** **entry_type=refund**, sabab majburiy, alohida ruxsat, chegaradan katta bo'lsa maker-checker

TUZATILDI — provayder protokollari aniqlashtirildi (v1.0 da “webhook + imzo” deb umumiy yozilgan, bu Payme uchun noto'g'ri):

Provayder	Protokol	Autentifikatsiya	Idempotency kaliti
Payme Merchant API	JSON-RPC 2.0: CheckPerformTransaction, CreateTransaction, PerformTransaction, CancelTransaction, CheckTransaction, GetStatement	HTTP Basic (merchant key)	Payme id (transaction)
Click SHOP API	Ikki bosqich: Prepare → Complete	sign_string (MD5) tekshiruvi	click_trans_id

- **SHART: UNIQUE** (**provider, provider_transaction_id**) — v1.0 dagi global unique turli provayderlarda ID to'qnashuvida xato beradi
- **SHART:** Provayder holat mashinasi (**created** → **performed** → **cancelled**) alohida **provider_transactions** jadvalida; ledger'ga faqat **performed** da yoziladi, **cancelled** da storno
- **SHART:** Autentifikatsiya xato bo'lsa — provayder protokolidagi xato kodi (Payme: -32504) + alert
- **SHART:** Kunlik reconciliation — **GetStatement** / Click hisoboti bilan solishtirish

M6.8 Hisobdan chiqarish (write-off)

Yaxlitlash qoldig'i, muddati o'tgan paket, bo'linmaydigan abonementning ishlatilmagan qismi → **entry_type=write_off**, alohida hisobot.

M6.9 Xarajatlar

Filial xarajatlari (ijara, kommunal, marketing, jihoz): toifa, summa, sana, izoh, chek fayli. **YANGI:** takrorlanuvchi xarajatlar (ijara har oy avtomatik qo'shiladi), marketing xarajatini reklama manbasiga bog'lash (CAC hisobi uchun).

M7. O'QITUVCHI MAOSHI (PAYROLL) — R2

M7.1 Hisoblash usullari

Usul	Kod	Tavsif
Fiksirlangan oylik	fixed_salary	Oyiga belgilangan summa
Soatbay (yagona)	hourly_flat	Har ak. soat uchun bir xil
Soatbay (differentsial)	hourly_per_unit	Har o'quv birligi uchun alohida
Suzuvchi stavka	hourly_by_attendance	Qatnashgan o'quvchi soniga qarab
Tushumdan foiz	revenue_percent	Yechim summasidan %
O'quvchi boshiga	per_student	Oyiga har faol o'quvchi uchun
Aralash	hybrid	Bir nechta usul yig'indisi

Suzuvchi stavka misoli: 0 o'quvchi → 30 000, 1 → 40 000, 2 → 50 000, 3+ → 60 000 so'm/soat. Hisobga kim kiradi — sozlanadi.

Qaror kutilmoqda: `revenue_percent` — **hisoblangan** yechim (accrual) dan yoki **tushgan** puldan (cash)? Qarzidor o'quvchi bo'lsa farq katta. Default taklif: accrual, sozlanadi.

M7.2 Jarayon

`payroll_period` → `calculate` → `payroll_entries` → `approve` → `pay`. Har bir `payroll_entry` da `calculation_snapshot JSONB` — **SHART**. O'qituvchi “nega kamroq oldim?” deb so'rsa, 3 oy oldingi stavka o'zgargan bo'lsa ham aniq javob beriladi.

M7.3 Qo'shimchalar va ushlanmalar

Bonus, jarima, avans, ushlanma — sabab va mas'ul majburiy. **YANGI:** KPI bonus qoidasi (masalan guruh retention ≥ 90% bo'lsa +X).

M7.4 Holatlar

`draft` → `calculated` → `approved` → `paid`; `approved` dan keyin qayta hisoblash bloklanadi, `reopen` audit'ga yoziladi.

M7.5 Almashtirish

Maosh `lesson.actual_teacher_id` ga hisoblanadi.

M7.6 O'qituvchi uchun shaffoflik (YANGI)

O'qituvchi kabinetida oy davomida **jonli taxminiy maosh** (o'tilgan darslar × stavka) va har bir dars bo'yicha tafsilot. Bu “nega kam” degan nizolarni oy oxirigacha emas, darhol hal qiladi.

M8. KOMMUNIKATSIYA

M8.1 Kanallar

Kanal	Provayder	Ustuvorlik	Izoh
Telegram bot	Telegram Bot API	1	Bepul, eng keng tarqalgan
SMS	Eskiz.uz / Play Mobile	2	Muhim xabarlar; shablon tasdig'i shart
Push	FCM / Web Push	3	PWA uchun Web Push (R1), FCM (R3)
Email	SMTP / Resend	4	Hujjatlar, hisobotlar

SHART — **kanal fallback** xabar turiga qarab: Telegram yetkazilmasa (bot bloklangan, `chat_id` yo'q) → SMS.

M8.2 Telegram'ni bog'lash (YANGI — v1.0 da yo'q edi)

Telegram bot `chat_id` ni telefon raqamidan o'zi bilmaydi. Oqim:

1. Tizim ota-onaga bir martalik havola yuboradi (SMS): `t.me/<bot>?start=<one_time_token>`
2. Bot `/start <token>` qabul qiladi → token tekshiriladi → `contact_person.telegram_chat_id` saqlanadi
3. Yoki botda “Raqamni ulashish” tugmasi (`request_contact`) — kelgan raqam normalizatsiya qilinib mos aloqa shaxsiga bog'lanadi
 - Token: 72 soat, bir martalik, hash holida saqlanadi
 - Bot bloklangan (403) — `telegram_blocked_at` belgilanadi, fallback SMS'ga o'tadi

M8.3 Shablonlar

O'zgaruvchilar (`{{student.name}}`, `{{balance}}`, `{{balance_lessons}}`, `{{payment_link}}`...), har kanal uchun variant, til `notification_language` bo'yicha (uz/ru/en). **SHART:** shablon saqlanayotganda o'zgaruvchilar validatsiya qilinadi; SMS uzunligi va segment soni ko'rsatiladi.

M8.4 Avtomatik xabarnomalar

Dars eslatmasi, bekor qilish, jadval o'zgarishi, schyot, muddat yaqin, qarzidorlik, to'lov qabul qilindi (chek bilan), farzand kelmadi, kurs tugashi, uy vazifasi, **oylik hisobot ota-onaga** (YANGI: davomat %, baholar, o'qituvchi izohi — bitta xabar).

SHART — anti-spam: kuniga maksimal xabar (default 5), jim vaqt 21:00–09:00 (kritiklardan tashqari, navbatda kutadi), har xabar `message_log` da (kanal, holat, xato, narx). **KERAK:** bir xil hodisa xabarlarini birlashtirish (digest).

M8.5 Ommaviy rassilka

Segment (saqlangan ko'rinish yoki filtr), preview, test yuborish, rejalashtirish, hisobot. **SHART:** yuborishdan oldin qabul qiluvchilar soni va SMS narxi ko'rsatiladi; 500+ qabul qiluvchi — tasdiqlash.

M8.6 Chatlar (R2)

Omnichannel inbox (Telegram → CRM), ichki chat, guruh chati, round-robin taqsimlash, SLA taymer, tez javob shablonlari.

M9. O'QUVCHI / OTA-ONA KABINETI (PWA)

M9.1 Autentifikatsiya

Telefon + SMS OTP; Telegram orqali kirish (deep link / Telegram Login); bitta ota-ona — bir nechta farzand (profil almashtirish).

SHART: OTP — 6 raqam, 5 daqiqa, 5 urinish, hash holida; OTP so'rovida raqam mavjudligi oshkor qilinmaydi (bir xil javob).

M9.2 Bo'limlar

Bosh sahifa (keyingi dars, balans so'm + **“yana N darsga yetadi”**, xabarlar), jadval, davomat, moliya (onlayn to'lov, chek yuklab olish), uy vazifalari, natijalar, chat, profil, **hujjatlar** (shartnomani ko'rish va OTP bilan tasdiqlash), **muzlatish so'rovi** (YANGI).

M9.3 Xavfsizlik

- **SHART:** `student_id` token/sessiyadan olinadi, URL'dan emas; ota-ona faqat `contact_person_students` orqali bog'langan farzandlarni ko'radi
- **SHART:** IDOR testlari CI'da

M9.4 O'qituvchi mobil ilovasi (PWA) — R1 (YANGI)

Alohida ixcham interfeys: bugungi darslar, bir bosishda davomat (offline), mavzu va uy vazifasi kiritish, jonli maosh, almashtirish so'rovi. Batafsil — UX v2.0 §5.

M10. HISOBOTLAR VA ANALITIKA — R3 (qarzdorlar R1)

M10.1 Moliyaviy

Tushumlar, yechimlar, to'lov statistikasi, **qarzdorlar (aging 0-30/31-60/61-90/90+)** — R1, avans qoldiqlari, sotuv rejasi, xarajatlar, rentabellik (filial/fan/guruh), operatsiyalar tarixi, **kassa smenalari** (YANGI), **cash-flow prognozi** (YANGI: kutilayotgan to'lovlar grafigi bo'yicha keyingi 30 kun).

M10.2 Sotuv

Voronka, status o'zgarishlari, manbalar (lead, konversiya, **CAC** — marketing xarajati / sotuv), ketish sabablari, menejer samaradorligi (javob vaqti, konversiya).

M10.3 Yuklama

Xonalar bandligi (heatmap: kun × soat), o'qituvchi yuklamasi, darslar statistikasi.

M10.4 KPI

KPI	Formula
Retention (oylik)	Davr oxirida qolganlar (yangilarsiz) / davr boshidagilar

KPI	Formula
Churn	Ketganlar / o'rtacha faol o'quvchilar
Kohort retention (YANGI)	Qo'shilgan oy bo'yicha N-oydagi qolganlar foizi
LTV	ARPU × o'rtacha o'qish davri (oy)
ARPU	Tushum / faol o'quvchilar
Konversiya	Won lead / jami lead
Guruh to'ldirilishi	O'quvchilar / capacity
Davomat %	Qatnashgan / bo'lgan darslar
Yig'ilish darajasi (YANGI)	Tushum / yechim (davr uchun)

SHART: Og'ir agregatsiya OLTP jadvallaridan emas: R1-R2 — materialized view + **REFRESH CONCURRENTLY** (har soat/kecha), R3 — read replica. Har hisobot Excel'ga eksport. Har bir KPI ta'rifi hisobot ichida "i" belgisida ko'rsatiladi (raqamga ishonch uchun).

M11. IMPORT / EKSPORT VA AUDIT

M11.1 Import

O'quvchilar, lead'lar, guruhlar, a'zoliklar, **boshlang'ich balanslar** (YANGI — migratsiya uchun kritik), to'lovlar.

Jarayon: fayl → mapping (saqlanadigan shablon) → dry-run validatsiya → xatolar Excel'i → tasdiq → import.

SHART: Har import `import_batch_id` bilan; bekor qilish (rollback) — import qilingan yozuvlarni arxivlash, ledger uchun storno. Katta fayl — background job, progress bilan.

M11.2 Eksport

CSV/XLSX; katta hajm — background job + havola. **SHART:** eksport ruxsat va filial scope'ga bo'ysunadi; kontakt maydonlari `*.view_contacts` siz yashiriladi.

M11.3 Audit log

Majburiy: moliyaviy operatsiyalar, rol/ruxsat o'zgarishi, kontakt ko'rish, retroaktiv davomat, payroll approve/reopen, ommaviy amallar, eksport, merge, super_admin support sessiyalari.

Format: `actor_id, action, entity_type, entity_id, diff JSONB, ip, user_agent, request_id, occurred_at`. **SHART:** append-only (trigger, M6.4 kabi), saqlash ≥ 3 yil, oylik partitsiya.

M12. YANGI FUNKSIYALAR (v2.0)

Bular referensda yo'q yoki zaif — raqobat ustunligi shu yerda.

M12.1 A'zolikni muzlatish (freeze) — R1

O'quvchi kasal, safarda yoki imtihonda — markazlar buni hozir qo'lda hal qiladi va pul noto'g'ri yechiladi. `enrollment_freezes(enrollment_id, from, to, reason, approved_by)`: oraliqdagi darslarda davomat avtomatik `frozen` (yechimsiz), `monthly` narxda pro-rata, sig'im band qilinmaydi (sozlanadi). Yilida maksimal muzlatish kunlari sozlanadi. Ota-ona kabinetdan so'rov yuboradi, administrator tasdiqlaydi.

M12.2 Navbat (waitlist) — R2

To'lgan guruhga yozilmoqchi bo'lganlar navbatga tushadi. Joy bo'shaganda (`enrollment` yopildi) `waitlist.seat_available` → birinchi navbatdagiga 24 soatlik taklif (Telegram/SMS), javob bo'lmasa keyingisiga. Yangi guruh ochish uchun ham signal: "B1 Backend, seshanba-payshanba 18:00 — navbatda 9 kishi".

M12.3 Oilaviy hisob (household) — R2

Bir ota-ona 2-3 farzand uchun to'laydi. `households` + `payer_client_id`: bitta to'lov bir nechta farzandga taqsimlanadi, kabinetda umumiy qarz, aka-uka chegirmasi avtomatik. Ledger har bir student uchun alohida qoladi (`transfer_in/out` bilan) — buxgalteriya aniq.

M12.4 Ketish xavfi (churn-risk) — R2

Qoida asosidagi ball (ML emas — tushunarli va tekshiriladigan): ketma-ket qoldirish (+30), oxirgi 30 kunda davomat < 60% (+25), 14+ kun qarz (+20), test bali pasaygan (+15), kurs oxiriga 14 kun qoldi va keyingi kursga yozilmagan (+10). Ball $\geq 50 \rightarrow$ "Xavf ostida" ro'yxati va menejerga vazifa. Har bir ball uchun **sabab ko'rsatiladi**. Bu G6 maqsadining asosiy vositasi.

M12.5 Bo'lib to'lash grafigi (installments) — R2

Paket yoki kurs narxi grafik bo'yicha bo'linadi (masalan $3 \times 1\,200\,000$). Har to'lov sanasi uchun avtomatik eslatma, kechikish $\rightarrow$ `installment.overdue`. Qarzdorlar hisobotida "grafik bo'yicha kechikkan" va "umumiy qarz" alohida.

M12.6 Kassa smenalari — R1

Naqd pul — suiiste'mol nuqtasi. Smena ochish/yopish, inkassatsiya, kutilgan va haqiqiy summa farqi. Batafsil M6.7.

M12.7 Sertifikat generatsiyasi — R3

Kurs tugaganda (minimal davomat va test bali sharti bilan) shablondan PDF sertifikat, QR-kod bilan **ochiq tekshirish sahifasi** (`/verify/<code>`). Bu bitiruvchilar uchun ham, markaz marketingi uchun ham qiymat.

M12.8 O'quv dasturi (curriculum) — R2

Kurs uchun mavzular ro'yxati (dars 1...N). Dars yakunlanganda o'qituvchi mavzuni belgilaydi; guruhning dasturdan **orqada qolishi** ko'rinadi. Ota-ona kabinetida "kurs 45% bajarildi". Mavzu bo'yicha uy vazifasi shabloni.

M12.9 Fikr-mulohaza (NPS) — R3

Har oy oxirida ota-ona/o'quvchiga Telegram'da 1 savol (0-10) + ixtiyoriy izoh. O'qituvchi va filial kesimida NPS. Past baho (≤ 6) $\rightarrow$ avtomatik vazifa.

M12.10 Public API va outgoing webhooks — R3

Tenant API kalitlari (scope bilan, hash holida), OpenAPI hujjati, outgoing webhook'lar (`student.enrolled`, `payment.received`...) HMAC imzo bilan, retry va yetkazish log'i. SaaS uchun integratsiya ekotizimining asosi.

M12.11 Ish stoli "xavf ostidagi pul" vidjeti — R1

Administrator har kuni ochganda: davomati belgilanmagan darslar (hisoblanmagan pul), bugun muddati tugaydigan schyotlar, 30+ kunlik qarzdorlar summasi. Har biri bir bosishda ro'yxatga olib boradi.

M12.12 AI yordamchi — R3 (ixtiyoriy)

Tor va tekshiriladigan vazifalar: qo'ng'iroq/chat matnidan lead izohini qisqartirish, ota-onaga oylik hisobot matnini qoralama qilish, tabiiy tildan filtr ("seshanba kunlari B1 guruhlardagi qarzdorlar").
SHART: Al hech qachon moliyaviy yozuv yaratmaydi va xabarni o'zi yubormaydi — faqat qoralama, xodim tasdiqlaydi.

5. MA'LUMOTLAR MODELI

5.1 Asosiy jadvallar (qisqartirilgan)

```
tenants(id, name, slug, timezone, currency, plan, is_active, created_at)
branches(id, tenant_id, name, code, address, phone, timezone, is_active)
classrooms(id, tenant_id, branch_id, name, capacity, is_active)

users(id, tenant_id, phone, email, password_hash, full_name, is_active,
      totp_secret_enc, last_login_at, created_at)
roles(id, tenant_id, code, name, is_system)
permissions(code, description)
role_permissions(role_id, permission_code, scope)
user_roles(user_id, role_id) user_branches(user_id, branch_id)
teacher_profiles(user_id, disciplines[], levels[], age_categories[], hired_at)
teacher_availability(id, teacher_id, day_of_week, time_from, time_to)
sessions(id, user_id, refresh_token_hash, family_id, expires_at, revoked_at)

households(id, tenant_id, name, primary_payer_id)
clients(id, tenant_id, household_id, full_name, birth_date, gender)
client_phones(id, client_id, phone_e164, is_primary) -- normalizatsiya
contact_persons(id, tenant_id, client_id, full_name, relation, phone_e164, email,
                is_bill_recipient, receives_notifications, telegram_chat_id, telegram_blocked_at)
contact_person_students(contact_person_id, student_id)

inbound_submissions(id, tenant_id, source, form_id, raw_payload jsonb,
                    utm_source, utm_medium, utm_campaign, utm_content, utm_term,
                    referrer, landing_url, ip, status, lead_id, created_at)
leads(id, tenant_id, client_id, branch_id, discipline_id, level_id, age_category_id,
      status_id, ad_source_id, learning_goal_id, owner_id, student_id,
      inbound_submission_id, lost_reason_id, tags text[], custom_data jsonb,
      created_at, first_contact_at, converted_at, version)
lead_status_history(id, lead_id, from_status_id, to_status_id, changed_by, changed_at)
lead_preferred_schedule(id, lead_id, day_of_week, time_from, time_to)

students(id, tenant_id, client_id, branch_id, status, owner_id,
          cached_balance bigint, churn_score smallint, blacklisted, blacklist_reason,
          custom_data jsonb, archived_at, merged_into_id, version)

study_units(id, tenant_id, branch_id, type, discipline_id, level_id, age_category_id,
            type_id, status, name, auto_name, capacity, min_size, default_price_id,
            responsible_id, color, curriculum_id, start_date, end_date, version)
enrollments(id, tenant_id, student_id, study_unit_id, price_id, discount_ids uuid[],
            payer_client_id, start_date, end_date, status, contract_id, version)
enrollment_freezes(id, enrollment_id, date_from, date_to, reason, status, approved_by)
waitlist_entries(id, tenant_id, study_unit_id, lead_id, student_id, position,
                 offered_at, offer_expires_at, status)

schedule_rules(id, tenant_id, study_unit_id, days_of_week int[], start_time, end_time,
```

```

    academic_hours_per_day, break_minutes, period_start, period_end,
    total_academic_hours, classroom_id, teacher_id, is_active)
lessons(id, tenant_id, study_unit_id, schedule_rule_id, date, start_at, end_at,
    classroom_id, teacher_id, actual_teacher_id, status, is_detached,
    academic_hours numeric(5,2), topic_id, topic, homework, notes,
    makeup_for_lesson_id, moved_to_lesson_id, cancel_reason, completed_at, completed_by)
attendances(id, tenant_id, lesson_id, student_id, enrollment_id, status, score,
    comment, marked_by, marked_at, client_marked_at) -- offline sync
UNIQUE (lesson_id, student_id)

prices(..., type, amount bigint, ..., superseded_by_id)
discounts(id, tenant_id, name, type, value bigint, reason, auto_rule jsonb,
    valid_from, valid_to)
invoices(id, tenant_id, student_id, enrollment_id, payer_client_id, number,
    issued_at, due_at, total_amount bigint, paid_amount bigint, status,
    pay_token_hash, created_by, version)
invoice_items(id, invoice_id, description, quantity, unit_amount, discount_amount, total_amount)
installment_plans(id, enrollment_id, total bigint)
installment_items(id, plan_id, due_date, amount bigint, invoice_id)

ledger_entries(... M6.4 ga qarang ...)
lesson_charges(lesson_id, student_id, current_entry_id, amount bigint, revision,
    PRIMARY KEY (lesson_id, student_id))
provider_transactions(id, tenant_id, provider, provider_transaction_id, invoice_id,
    amount bigint, state, raw jsonb, created_at, performed_at, cancelled_at,
    UNIQUE (provider, provider_transaction_id))
cash_shifts(id, tenant_id, branch_id, opened_by, opened_at, closed_at,
    expected_cash bigint, actual_cash bigint, diff_comment)
document_sequences(tenant_id, doc_type, year, last_value, PRIMARY KEY(...))

teacher_rates(id, tenant_id, teacher_id, study_unit_id, method, amount bigint,
    attendance_bands jsonb, valid_from, valid_to)
payroll_periods(id, tenant_id, year, month, status, calculated_at, approved_at,
    approved_by, paid_at)
payroll_entries(id, tenant_id, period_id, teacher_id, source_type, source_id,
    hours numeric, rate bigint, amount bigint, calculation_snapshot jsonb)
payroll_adjustments(id, tenant_id, period_id, teacher_id, type, amount bigint,
    reason, created_by)

message_templates, notification_rules, message_log, automations,
automation_runs, outbox_events(..., causation_id, causation_depth),
tasks, expenses(..., recurring_rule, ad_source_id), audit_log (oylik partitsiya),
custom_field_definitions, saved_views(id, user_id, entity, name, filters jsonb,
    columns jsonb, sort, is_default, is_shared),
idempotency_keys(tenant_id, key, request_hash, response jsonb, created_at),

```

```

curricula, curriculum_topics, certificates, nps_responses, api_keys, webhook_subscriptions

```

5.2 Index strategiyasi (boshlang'ich)

```

CREATE INDEX ON students (tenant_id, branch_id, status);
CREATE INDEX ON students (tenant_id, cached_balance) WHERE cached_balance < 0;
CREATE INDEX ON students (tenant_id, churn_score DESC) WHERE churn_score >= 50;
CREATE INDEX ON leads (tenant_id, status_id, created_at DESC);
CREATE INDEX ON leads (tenant_id, owner_id, status_id);
CREATE INDEX ON leads (tenant_id, created_at) WHERE first_contact_at IS NULL; -- SLA
CREATE INDEX ON lessons (tenant_id, study_unit_id, start_at);
CREATE INDEX ON lessons (tenant_id, actual_teacher_id, start_at);
CREATE INDEX ON lessons (tenant_id, classroom_id, start_at);
CREATE INDEX ON lessons (tenant_id, date) WHERE status = 'planned';
CREATE INDEX ON attendances (student_id, marked_at DESC);
CREATE INDEX ON ledger_entries (tenant_id, student_id, occurred_at DESC);
CREATE INDEX ON ledger_entries (tenant_id, occurred_at) WHERE entry_type = 'payment';
CREATE UNIQUE INDEX ON ledger_entries (tenant_id, idempotency_key)
    WHERE idempotency_key IS NOT NULL;
CREATE INDEX ON client_phones (phone_e164);
CREATE INDEX ON clients USING gin (full_name gin_trgm_ops); -- pg_trgm: qisman moslik
CREATE INDEX ON outbox_events (next_retry_at) WHERE status = 'pending';

```

TUZATILDI. `to_tsvector('simple', full_name)` ism qidiruvi uchun yomon tanlov — “Muham” bilan “Muhammad” topilmaydi, o'zbekcha '/' variantlari mos kelmaydi. `pg_trgm + unaccent` va apostrof normalizatsiyasi (' ' ' ' → ') ishlatiladi.

Qoida: qolgan index'lar `pg_stat_statements` tahlilidan keyin.

5.3 Multi-tenancy — Row-Level Security

```
ALTER TABLE students ENABLE ROW LEVEL SECURITY;
ALTER TABLE students FORCE ROW LEVEL SECURITY;           -- jadval egasiga ham
CREATE POLICY tenant_isolation ON students
  USING (tenant_id = current_setting('app.current_tenant', true)::uuid)
  WITH CHECK (tenant_id = current_setting('app.current_tenant', true)::uuid);
```

TUZATILDI — kritik. v1.0: “ORM middleware har so'rov boshida `SET LOCAL` bajaradi”. Lekin:

- 1) `SET LOCAL` faqat **transaction** ichida amal qiladi; Prisma oddiy query'larni alohida pool ulanishlarida bajaradi — sozlama yo'qoladi yoki boshqa so'rovga “oqib” o'tadi.
- 2) Migratsiyalar egasi bo'lgan rol bilan ishlasa RLS umuman qo'llanmaydi (`FORCE` siz).
- 3) `WITH CHECK` siz boshqa tenant'ga `INSERT` qilish mumkin.

Yechim: har so'rov `prisma.$transaction` ichida `SELECT set_config('app.current_tenant', $1, true)` bilan boshlanadi (Prisma Client extension); ilova `app_user` roli bilan ulanadi (egasi emas, `BYPASSRLS` yo'q); migratsiya — alohida `migrator` roli; PgBouncer **transaction mode**. CI'da test: tenant konteksti o'rnatilmagan so'rov 0 qator qaytaradi.

6. API SPETSIFIKATSIYASI

6.1 Konventsiyalar

- Base: `/api/v1`, JSON, ISO 8601 UTC sanalar, pul — tiyinda **string** sifatida (`"amount": "50000000"`) — JS **number** 2^{53} dan katta summada anqlik yo'qotadi va pul bigint ekanini kontraktida aniqlaydi
- Autentifikatsiya: admin web — httpOnly cookie (BFF orqali); mobil/integratsiya — **Authorization: Bearer**
- Pagination: cursor (`?limit=50&cursor=...`), sort `?sort=-created_at`, filtr `?filter[status]=active`
- **SHART:** OpenAPI 3.1 sxemasi koddan generatsiya qilinadi; frontend tiplari undan avtomatik (`openapi-typescript`) — qo'lda yozilgan tip yo'q
- **SHART:** Optimistic locking — `PATCH` so'rovlari **If-Match: <version>** qabul qiladi; mos kelmasa **409 VERSION_CONFLICT**

6.2 Javob va xato formati

```
{ "data": { }, "meta": { "next_cursor": "...", "total": 1234 } }

{ "error": {
  "code": "SCHEDULE_CONFLICT",
  "message": "Xona bu vaqtda band",
  "details": [{ "field": "classroom_id", "code": "BUSY",
    "conflict_lesson_id": "...",
    "suggestions": [{ "classroom_id": "...", "start_at": "..."} ] }],
  "request_id": "01JBQ..." } }
```

SHART: Har javobda **X-Request-Id**; log'larda ham. Xato kodlari ro'yxati (enum) hujjatlashtiriladi — frontend **message** matniga emas, **code** ga tayanadi (i18n uchun).

6.3 Asosiy endpoint'lar

Auth	POST /auth/login /auth/refresh /auth/logout /auth/otp/request POST /auth/otp/verify /auth/2fa/verify GET /auth/me
Inbound	POST /public/inbound GET /inbound POST /inbound/:id/process
Leads	GET POST /leads GET PATCH /leads/:id POST /leads/:id/status POST /leads/:id/convert /leads/:id/trial /leads/bulk
Students	GET POST /students GET PATCH /students/:id POST /students/:id/archive /students/:id/merge /merges/:id/undo GET /students/:id/ledger /attendance /schedule /timeline POST /students/:id/documents
Units	GET POST /study-units GET PATCH /study-units/:id POST /study-units/:id/status GET POST /study-units/:id/members POST /study-units/:id/transfer POST /enrollments/:id/freeze GET POST /study-units/:id/waitlist
Schedule	POST /study-units/:id/schedule-rules PATCH /schedule-rules/:id POST /schedule-rules/:id/preview GET /lessons PATCH /lessons/:id POST /lessons/:id/cancel move complete POST /lessons/bulk-cancel GET PUT /lessons/:id/attendance GET /schedule/free-classrooms /schedule/free-teachers
Finance	GET POST /prices GET POST /invoices POST /invoices/bulk /invoices/:id/cancel GET POST /payments POST /payments/:id/refund POST /ledger/adjustments POST /cash-shifts/open /cash-shifts/:id/close POST /installment-plans GET /reports/debtors?aging=true POST /payments/providers/payme -- JSON-RPC, Basic auth POST /payments/providers/click/prepare /click/complete
Payroll	GET /payroll/periods POST /payroll/periods/:id/calculate approve pay reopen GET /payroll/periods/:id/entries POST /payroll/adjustments
Messages	GET POST /templates GET PATCH /notification-rules/:id POST /messages/send /messages/bulk GET /messages/log GET POST /automations POST /automations/:id/dry-run
Views	GET POST PATCH DELETE /saved-views
Portal	GET /portal/me /portal/children /portal/schedule /portal/attendance GET /portal/ledger /portal/invoices POST /portal/invoices/:id/pay GET /portal/homework POST /portal/homework/:id/submit POST /portal/freeze-requests POST /portal/documents/:id/sign
Teacher	GET /teacher/today PUT /teacher/lessons/:id/attendance (offline batch) GET /teacher/earnings
Public	GET /p/:pay_token GET /verify/:certificate_code

6.4 Idempotency

SHART: Pul bilan bog'liq barcha POST'lar **Idempotency-Key** qabul qiladi. Saqlash:

`idempotency_keys(tenant_id, key, request_hash, response, created_at)`, TTL 24 soat. Bir xil kalit + **boshqa request_hash** → 422 **IDEMPOTENCY_KEY_REUSED**. Parallel bir xil kalit → ikkinchisi 409 (kalit qatori **INSERT ... ON CONFLICT** bilan band qilinadi).

6.5 Rate limiting

Guruh	Limit
/auth/login, /auth/otp/request	5 / 15 daq / IP va akkaunt
/public/inbound	10 / soat / IP + captcha
Umumiy autentifikatsiyalangan	300 / daq / foydalanuvchi
Eksport / hisobot	10 / daq / foydalanuvchi
Public API (kalit bilan)	600 / daq / kalit

7. ARXITEKTURA

7.1 Modulli monolit

```

Next.js (App Router): Admin CRM | PWA: ota-ona kabineti | PWA: o'qituvchi
                        \_____ BFF (Next route handlers, cookie) ____/
                                | REST /api/v1

NestJS modulli monolit:
  iam | crm | academy | finance | payroll | messaging | reporting | integrations
  shared: audit, outbox, files, money, tenancy, idempotency
    |
  PostgreSQL 16      Redis 7 (cache, queue, S3 (MinIO/obyekt)
  (+ PgBouncer)      rate limit)
    |
  BullMQ workers: outbox, notifications, schedule-gen, charge, reports,
                  reconciliation, imports

```

Nega mikroservis emas? Jamoa 2-5 kishi; domenlar kuchli bog'langan (dars → yechim → balans → xabar) — taqsimlash distributed transaction (saga) muammosini keltiradi. Modulli monolit chegaralari toza bo'lsa keyin bo'linadi.

SHART — modul chegaralari: modul boshqa modul repository/entity'siga murojaat qilmaydi — faqat public service interface yoki domain event. CI'da **dependency-cruiser**.

KERAK — worker ajratish: API va worker bitta kod bazasi, lekin alohida process (**APP_ROLE=api|worker**) — og'ir hisobot API latency'ga ta'sir qilmaydi.

7.2 Texnologiyalar

Qatlam	Tanlov	Asos
Backend	NestJS 11 + TypeScript strict	Modul tizimi domen chegaralarini ushlaydi; DI
ORM	Prisma (+ \$queryRaw hisobotlar uchun)	Type-safe, migratsiya. Exclusion constraint, trigger, RLS — SQL migratsiyada
DB	PostgreSQL 16 + PgBouncer	Exclusion constraint, RLS, JSONB, window functions
Queue	Redis 7 + BullMQ	Job, rate limit, kesh
Frontend	Next.js 15 + TypeScript	Admin uchun asosan client component; SSR faqat ommaviy sahifalarda
UI	Tailwind + shadcn/ui + TanStack Table (virtualizatsiya)	Katta jadvallar
State	TanStack Query	Server state, optimistic update
Formalar	React Hook Form + Zod	Zod sxemalari OpenAPI'dan
PWA	Servist (service worker) + IndexedDB	Offline davomat
PDF	Gotenberg (HTML → PDF)	Shartnoma, chek, sertifikat
Fayl	S3-compatible	
Monitorin g	Sentry + Prometheus + Grafana + Loki (Pino)	
Feature flags	Unleash (self-hosted) yoki DB-flag	Bosqichma-bosqich yoqish

TUZATILDI. v1.0 da frontend uchun asos “SSR (SEO kerak sahifalar), RSC bilan tez ro'yxatlar” edi. Admin CRM — login ortida, SEO kerak emas; filtrli interaktiv jadvallar client-side kesh bilan tezroq. RSC faqat layout va ommaviy sahifalarda.

SHART: **strict: true, noUncheckedIndexedAccess: true; any** — code review'da asoslanadi.

7.3 Kritik pattern'lar

- Outbox** — biznes o'zgarishi va tashqi ta'sir bir transaction'da bo'lmaydi; hodisa outbox'ga yoziladi, worker yuboradi.
- Domain events** — **lesson.completed** → **finance (yechim), messaging, payroll**. Har handler mustaqil va idempotent.

3. **Optimistic locking** — **version** ustuni (study_unit, enrollment, invoice, lead, student).
4. **Arxivlash, soft delete emas** — **deleted_at** o'rniga biznes status.
5. **Advisory lock** (YANGI) — bitta o'quv birligi uchun jadval regeneratsiyasi va charge hisoblash parallel ishlamasligi uchun **pg_advisory_xact_lock(hash(study_unit_id))**.
6. **Money value object** (YANGI) — barcha pul arifmetikasi bitta joyda va test qilingan.

8. NOFUNKSIONAL TALABLAR

8.1 Unumdorlik

Ko'rsatkich	Maqsad
API p95 / p99	< 300 ms / < 800 ms
Jadval (oylik ko'rinish)	< 1 s
Davomat saqlash (o'qituvchi)	< 500 ms, offline'da darhol
Hisobot	< 5 s interaktiv, aks holda background
Bir vaqtdagi foydalanuvchi	200 (R1), 2000 (R3)
Hajm	50 000 o'quvchi, 5 mln dars, 20 mln ledger
Frontend	LCP < 2.5 s, INP < 200 ms; initial JS < 250 KB gz

SHART: N+1 taqiqlanadi; CI'da query count testlari.

8.2 Ishonchlilik

Uptime 99.5%, RPO 15 daq (WAL archiving, pgBackRest), RTO 2 soat. Kunlik to'liq + uzluksiz WAL; **haftalik avtomatik restore testi** (test qilinmagan backup — backup emas). Backup boshqa jismoniy joyda.

8.3 Xavfsizlik

Autentifikatsiya va sessiya

- Parol: Argon2id, min 10 belgi, sizib chiqqan parollar ro'yxatiga tekshiruv
- Access token 15 daq; refresh 30 kun, rotation + reuse detection (**family_id**)
- **TUZATILDI:** Admin web'da token'lar JavaScript'ga ochiq saqlanmaydi — httpOnly, Secure, SameSite=Lax cookie (BFF). localStorage'dagi JWT — XSS bilan bevosita o'g'irlanadi
- 2FA (TOTP) — owner, accountant uchun majburiy; boshqalar uchun ixtiyoriy
- Aktiv sessiyalar ro'yxati va "barcha qurilmalardan chiqish"

Avtorizatsiya: guard (M3.3), IDOR testlari CI'da, super_admin support sessiyasi vaqt bilan cheklangan va audit'da.

Ma'lumot himoyasi: TLS 1.3 + HSTS, disk va backup shifrlangan, passport ma'lumoti application-level shifrlash (envelope encryption, kalit secret manager'da), sirlar repozitoriyda emas (gitleaks pre-commit + CI).

Input: DTO validatsiya, faqat parametrlangan SQL, **dangerouslySetInnerHTML** taqiqlangan (lint), fayl — MIME + magic-byte + hajm + alohida domendan serve, CSP header'lar, CSRF — SameSite + double-submit token (BFF).

Monitoring: 5 muvaffaqiyatsiz login → vaqtincha bloklash; anomaliya alert (ommaviy eksport, tungi moliyaviy operatsiya, ko'p kontakt ko'rish).

8.4 Shaxsiy ma'lumotlar va huquqiy jihat

Yurist xulosasi SHART (development boshlanishidan oldin). “Shaxsga doir ma’lumotlar to’g’risida”gi qonun O’zbekiston fuqarolarining shaxsiy ma’lumotlarini O’zbekiston hududidagi serverlarda saqlashni talab qiladi; tizimda voyaga yetmaganlar ma’lumoti ham bor. Amaliy natija: hosting O’zbekistonda (yoki lokalizatsiya talabi bajarilgan), operator sifatida ro’yxatdan o’tish ehtimoli, rozilik mexanizmi, voyaga yetmaganlar uchun ota-onalar roziligi. Bu hosting tanloviga ta’sir qiladi.

Texnik ta’minot: rozilik yozuvlari (**consents**: kim, qachon, qaysi matn versiyasi), ma’lumot eksporti so’rov bo’yicha, o’chirish so’rovi (moliyaviy yozuvlar saqlanadi, shaxsiy maydonlar anonimizatsiya), kontakt ko’rish log’i, saqlash muddati siyosati.

8.5 Lokalizatsiya

O’zbek (lotin) birlamchi, rus, ingliz; **DD.MM.YYYY; 1 234 567 so’m; Asia/Tashkent**; DB’da **timestamp** UTC. **SHART:** UI matnlari kodga yozilmaydi — i18n kalitlari; tarjima yo’qligi CI’da tekshiriladi. Apostrof normalizatsiyasi (o’, g’) qidiruv va saralashda.

8.6 Foydalanish qulayligi

WCAG 2.1 AA; klaviatura navigatsiyasi va shortcut’lar; admin 1280px+ (planshet 1024px — o’qish rejimi), kabinet va o’qituvchi — mobile-first. Batafsil — UX v2.0.

9. INTEGRATSIYALAR

Integratsiya	Provayder	Release	Izoh
To’lov	Payme	R1	Merchant API (JSON-RPC)
To’lov	Click	R1	SHOP API (Prepare/Complete)
To’lov	Uzum Bank	R3	
SMS	Eskiz.uz	R1	Shablon tasdig’i 1-2 hafta — erta boshlash
SMS	Play Mobile	R2	Fallback
Messenger	Telegram Bot API	R1	Xabar, kirish, chat (R2)
Push	Web Push / FCM	R1 / R3	
Email	SMTP / Resend	R1	
Telefoniya	Sipuni / OnlinePBX	R3	Qo’ng’iroq yozuvi lead’ga
Forma	O’z embeddable widget	R2	UTM bilan
Kalendar	Google Calendar (ICS feed)	R3	O’qituvchi jadvali — faqat o’qish uchun ICS

SHART — provayder abstraksiyasi: **PaymentProvider**, **SmsProvider**, **MessengerProvider** interfeyslari; SDK biznes logikaga kirmaydi. Har bir adapter uchun **sandbox contract test**.

SHART — degradation: tashqi servis ishlamasa tizim ishlaydi; xabarlar navbatda, circuit breaker, to’lov qabuli bloklanmaydi.

10. TESTLASH STRATEGIYASI

10.1 Darajalar

Daraja	Qamrov	Vosita
Unit	Narx, balans, payroll, generatsiya, Money	Vitest
Property-based	Ledger invariantlari, pro-rata, chegirma	fast-check
Integration	Repository + DB, constraint, RLS, trigger	Vitest + Testcontainers
API e2e	Endpoint, ruxsat, validatsiya	Supertest
Contract	Payme/Click/Eskiz/Telegram adapterlari	Sandbox + yozib olingan javoblar

Daraja	Qamrov	Vosita
UI e2e	Kritik yo'llar (F1-F4)	Playwright
Yuklama	200 foydalanuvchi, jadval va davomat	k6
Security	IDOR, permission bypass, injection	Avtomatik + pentest R1 oldidan

10.2 Qamrov

Branch coverage $\geq 85\%$: yechim dvigateli, balans, payroll, jadval generatsiyasi va konflikt, ruxsatlar, provayder adapterlari. Qolgani $\geq 60\%$. **Mutation testing** (Stryker) moliya moduli uchun — coverage “kod ishladi” deydi, mutation esa “test xatoni ushladi” deydi.

10.3 Majburiy stsensariylar

```

FINANCE
✓ Dars yakunlandi → har qatnashganga yechim
✓ Dars ikki marta yakunlandi → yechim ikkilanmaydi
✓ present → absent_excused → present: ledger -50k, +50k, -50k; lesson_charges.revision = 3
✓ Dars bekor qilindi → storno, asl yozuv o'zgarmaydi
✓ ledger_entries UPDATE/DELETE → EXCEPTION (jimgina o'tib ketmaydi)
✓ SUM(ledger) == cached_balance (property-based, 10 000 tasodifiy operatsiya)
✓ monthly pro-rata: oy o'rtasida qo'shilish, bayram, muzlatish
✓ Chegirmalar yig'indisi narxdan oshmaydi
✓ Payme CreateTransaction ikki marta → bitta tranzaksiya; CancelTransaction → storno
✓ Click Prepare noto'g'ri sign_string → rad etiladi
✓ Muzlatilgan davrdagi darsga yechim yo'q
SCHEDULE
✓ Kesishuvchi ikki dars bir xonada / bir o'qituvchida → constraint xatosi
✓ Ko'chirilgan (moved) darsning eski vaqti bo'sh hisoblanadi
✓ Parallel ikki so'rov bir xonaga → biri SCHEDULE_CONFLICT (suggestions bilan)
✓ Qoida o'zgarganda o'tgan va is_detached darslar tegilmaydi
✓ Bayram kuni dars generatsiya qilinmaydi
TENANCY & PERMISSIONS
✓ Tenant konteksti o'rnatilmagan so'rov → 0 qator
✓ Tenant A ma'lumotiga tenant B tokeni bilan → 404
✓ teacher boshqa o'qituvchi maoshini ko'ra olmaydi
✓ Ota-ona boshqa bolaning ledger'ini ko'ra olmaydi (IDOR)
✓ Guard'siz route → CI build xatosi
AUTOMATION
✓ A→B→A tsikl → depth 3 da to'xtaydi
✓ Jim vaqtda yuborilgan oddiy xabar → 09:00 gacha navbatda
PAYROLL
✓ Suzuvchi stavka band to'g'ri; almashtirishda actual_teacher_id ga
✓ approved davrni qayta hisoblash bloklangan; snapshot to'liq
OFFLINE
✓ O'qituvchi offline'da davomat qo'ydi → online'da sinxronlandi, dublikat yo'q

```

11. DEPLOYMENT VA INFRATUZILMA

11.1 Muhitlar

local (Docker Compose, seed), staging (anonimlashtirilgan prod nusxasi), production. **YANGI:** PR uchun preview muhit (ixtiyoriy, R2+).

11.2 CI/CD

```

push → lint + typecheck + i18n check + route-guard check + dependency-cruiser
→ unit → build → integration (Testcontainers) → security (npm audit, gitleaks, Trivy)
→ staging deploy → migrations → e2e (Playwright) → manual approval → production

```

SHART — migratsiyalar: deploy'dan oldin alohida; backward-compatible (expand → migrate → contract); ustun o'chirish ≥ 2 release; katta jadvalda **CREATE INDEX CONCURRENTLY**; **lock_timeout = 5s**; har birida rollback rejasi.

11.3 Deployment

Docker + Compose (R1) → Kubernetes kerak bo'lsa. Zero-downtime: rolling update, graceful shutdown (SIGTERM → yangi so'rov yo'q → 30 s). `/health/live`, `/health/ready`. Feature flag'lar. Hosting — O'zbekistondagi data-markaz (8.4 ga qarang).

11.4 Monitoring va alert

Metrikalar: API (RPS, latency, xato %), DB (pool, sekin query, lock'lar), queue (uzunlik, dead-letter), **biznes** (kunlik to'lovlar, yuborilmagan xabarlar, davomati belgilanmagan darslar, muvaffaqiyatsiz auto-action'lar).

Darhol alert: xato > 1% (5 daq), **ledger reconciliation nomuvofiqligi (kritik)**, **provayder reconciliation farqi (kritik)**, queue > 1000, webhook xatolari, backup bajarilmadi, SSL muddati < 14 kun.

12. ROADMAP

Bosqich	Mazmun	Davom iylik	Release
0. Poydevor	Skelet, auth + 2FA, RBAC, tenancy/RLS, filial, xodim, CI/CD, monitoring, Money, outbox	3 hafta	R1
1. Akademik yadro	Ma'lumotnomalar, o'quvchi, o'quv birligi, jadval + konflikt, davomat, o'qituvchi PWA	5 hafta	R1
2. Moliya	Narxlar, ledger, yechim dvigateli, schyot, to'lov, kassa smenasi, Payme/Click, qarzdorlar, muzlatish	5 hafta	R1
3. Kabinet + xabarlar	Ota-ona PWA, Telegram bog'lash, asosiy xabarnomalar, import	3 hafta	R1
4. Pilot	1 filial, parallel ishlash, tuzatishlar	4 hafta	R1 tayyor (~20 hafta)
5. CRM	Inbound/UTM, lead, voronka, sinov, vazifa, auto-actions	4 hafta	R2
6. Retention	Payroll, churn-risk, navbat, oilaviy hisob, bo'lib to'lash, rassilka, chat	5 hafta	R2 tayyor (~29 hafta)
7. Analitika va SaaS	Hisobotlar, KPI, sertifikat, NPS, public API, onboarding	5 hafta	R3 (~34 hafta)

Jamoa: 2-3 fullstack + dizayner + 0.5 QA. Har bosqichga **20% bufer** kiritilgan.

Asosiy tavsiya: R1 — bu “pul to'g'ri hisoblanadigan, administrator Excel'dan voz kechadigan” tizim. Undan oldin lead/voronka qilish — chiroyli, lekin pul oqmayotgan joyni avtomatlashtirish. R1 real ishlatilgandan keyin R2 ustuvorliklari qayta ko'rib chiqiladi.

12.1 Migratsiya rejasi

1. Mavjud ma'lumot eksporti va tahlili
2. Mapping hujjati (eski → yangi maydon)
3. Staging'da test import, nomuvofiqliklar hisoboti
4. **Boshlang'ich balanslar adjustment** (sabab: “migratsiya”) bilan, har o'quvchi uchun eski tizim bilan imzolangan solishtirish
5. Pilot filial — 1 oy parallel ishlash
6. Balans, davomat, maosh — 100% moslik; keyin boshqa filiallar

13. TAYYORLIK MEZONLARI

13.1 Definition of Done

- Kod yozilgan, self-review; TypeScript strict, asoslanmagan **any** yo'q
- Unit + integration testlar o'tgan; moliya o'zgarishida property-based test
- OpenAPI yangilangan; ruxsat guard + test; kerak bo'lsa audit log
- i18n kalitlari (uz/ru/en) qo'shilgan
- Migration backward-compatible, rollback rejasi bor
- Code review (≥ 1 tasdiq; moliya modulida ≥ 2)
- Staging'da tekshirilgan; UX: bo'sh, yuklanish va xato holatlari bor

13.2 R1 acceptance criteria

- Administrator guruh yaratadi, jadval qo'yadi, 10 o'quvchi qo'shadi — **5 daqiqada**
- O'qituvchi telefondan 15 kishilik guruh davomatini **20 soniyada** belgilaydi (offline ham)
- Xona/o'qituvchi konflikti DB darajasida oldi olinadi, muqobil taklif bilan
- Yechim barcha narx modellari (monthly pro-rata, muzlatish bilan) uchun avtomatik va to'g'ri
- Balans har doim **SUM(ledger)** ga teng — reconciliation 30 kun xatosiz
- Ota-ona Telegram'ni ulaydi, balans va jadvalni ko'radi, Payme/Click orqali to'laydi; chek avtomatik keladi
- Qarzdorlik eslatmasi jim vaqtga rioya qilib avtomatik yuboriladi
- Kassa smenasi yopilganda farq ko'rsatiladi
- Migratsiya: balanslar 100% mos
- Yuklama: 200 foydalanuvchi, p95 < 300 ms; pentest'da kritik/yuqori topilma yo'q

14. RISKLAR VA OCHIQ SAVOLLAR

14.1 Risklar

#	Risk	Ehti moli k	Ta'sir	Chora
R1	Moliyaviy xato (noto'g'ri balans, ikkilangan yechim)	O'rta	Juda yuqori	Append-only ledger, lesson_charges , reconciliation, property + mutation test, parallel pilot
R2	Scope creep	Yuqori	Yuqori	Release modeli; har talabga "qaysi biznes muammoni yechadi va qaysi release?"
R3	Shaxsiy ma'lumotlar qonunchiligi	O'rta	Yuqori	Yurist xulosasi development'dan oldin
R4	Migratsiya	Yuqori	Yuqori	Dry-run, boshlang'ich balans importi, 1 oy parallel
R5	To'lov provayderlari (sandbox, hujjat)	O'rta	O'rta	Erta boshlash, contract test, abstraksiya
R6	Jadval murakkabligi	Yuqori	O'rta	Bosqich 1 da real jadval ma'lumotida prototip
R7	Administratorlar Excel'ga o'rgangan	O'rta	O'rta	Pilot, o'qitish, klaviatura-first UX, Excel import/eksport
R8	Jamoa yetarli emas	O'rta	Yuqori	Release'lar, 20% bufer
R9	O'qituvchilar davomatni o'z vaqtida qo'ymaydi (YANGI)	Yuqori	Yuqori	Mobil PWA, eslatmalar, "xavf ostidagi pul" vidjeti, KPI
R10	SMS xarajati kutilgandan yuqori (YANGI)	O'rta	O'rta	Telegram-first, xarajat log'i, oylik limit va alert

14.2 Taxminlar

Jamoa: 2-3 fullstack + dizayner + 0.5 QA; hosting O'zbekistonda; Payme/Click merchant akkauntlari; Eskiz shablonlari tasdiqlanadi; pilot uchun bitta filial.

14.3 Ochiq savollar (development'dan oldin yopilishi kerak)

1. Yurist xulosasi: hosting, rozilik, voyaga yetmaganlar
2. Najot Ta'limdagi narx modeli: oylik / kurs / dars? Oy o'rtasida qo'shilish qanday hisoblanadi?
3. O'qituvchi maoshi formulasi; **revenue_percent** — accrual yoki cash?
4. Filial, o'quvchi, o'qituvchi soni (yuklama baholash)
5. Mavjud ma'lumot qayerda va qaysi formatda?
6. Sinov darsi siyosati: bepulmi, nechta?
7. Qarzdorlik siyosati: qancha qarzda darsga qo'yilmaydi? (Tizim bloklashi kerakmi yoki faqat ogohlantirish?)
8. Sababli qoldirish (absent_excused) qanday hisoblanadi — to'liq / qisman / yo'q?
9. Muzlatish siyosati: yiliga necha kun, kim tasdiqlaydi?
10. SaaS'ga chiqish rejasi qachon?

15. HUQUQIY VA ETIK ESLATMA

TZ referens mahsulotning **ochiq hujjatlari** asosida funksional talablarni shakllantiradi. Taqiqlanadi: referens kodini olish, UI dizayni/ikonka/rang/tipografiyani ko'chirish, matn va shablonlarni ko'chirish, brend elementlaridan foydalanish, referens akkauntiga kirib uni tizimli hujjatlashtirish (odatda ToS buzilishi).

Ruxsat etilgan: funksionallik, biznes-jarayon, domen modeli — ochiq domen bilimi.

Strategik tavsiya: mahsulotni klon emas, **O'zbekiston bozori uchun mustaqil mahsulot** sifatida pozitsiyalash: Payme/Click, Telegram-first, o'zbek tili, lokal qonunchilik, oylik abonement modeli, o'qituvchi uchun mobil tajriba — referens bera olmaydigan ustunliklar.

Hujjat oxiri. TZ v2.0 — 07.10.2026.